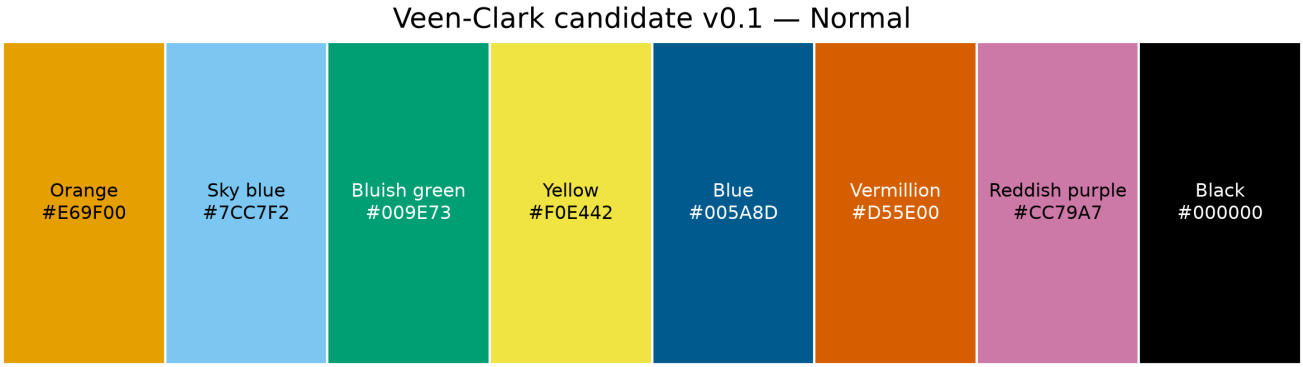
Veen-Clark palette

Complete candidate v0.1 computational audit



Candidate palette in normal sRGB.

An experimental Okabe-Ito derivative by Roland J. Veen and Charlie Clark. Sky blue is lightened and blue is darkened so the pair can represent related subgroups in scientific scatter and bar charts.

Status: candidate. The report presents transparent computational evidence, not a universal accessibility certification or participant validation.

Generated with PaletteBench 0.1.0 using Python 3.12.14. Exact inputs, hashes, dependencies, full-precision CSV/JSON data, and individual vector figures accompany this PDF in the repository.

Executive findings

The intended sky-blue/blue subgroup pair gains substantial modelled separation relative to Okabe-Ito under every standard reported condition: normal, protan 100%, deutan 100%, tritan 100%, and grayscale.

Condition	Okabe-Ito	Candidate	Change
Normal	22.31	37.92	+15.61
Protan 100%	20.75	36.03	+15.28
Deutan 100%	22.45	38.57	+16.12
Tritan 100%	19.95	35.08	+15.13
Grayscale	21.62	37.43	+15.81

The whole-palette result is mixed. The normal-vision 10th percentile rises by 2.73 Delta E00 and the protan-100% minimum rises by 2.92, but darkening blue reduces separation from black and produces the largest loss against vermillion under tritan 100% (-12.20 Delta E00).

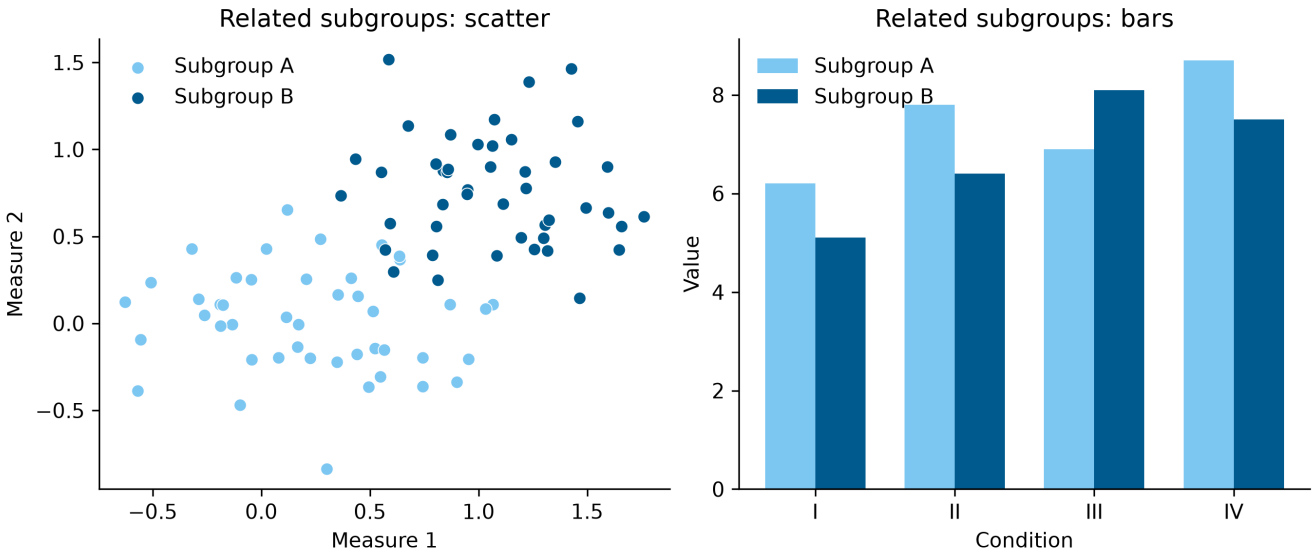


Figure 1. Related blue subgroups shown with colour as the only distinguishing visual encoding.

Baseline-relative comparison

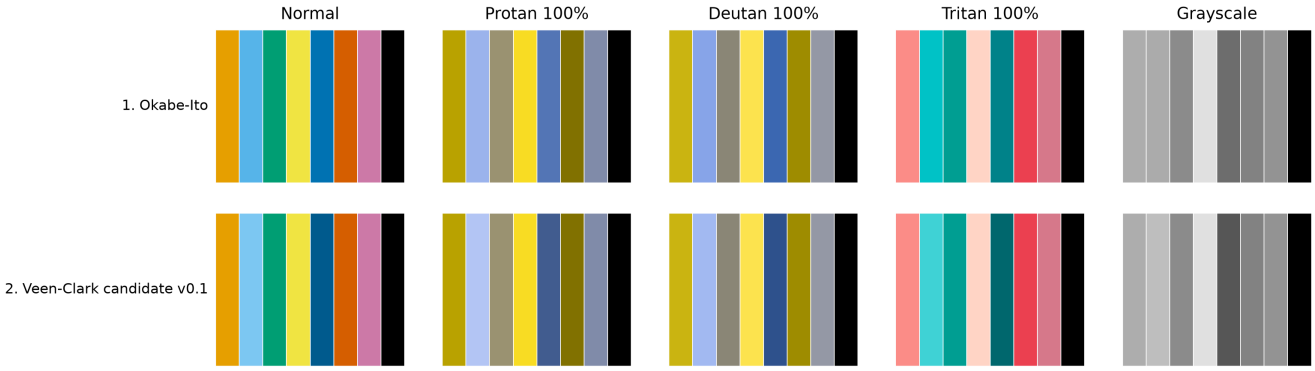


Figure 2. Okabe-Ito baseline and Veen-Clark candidate under the standard reported conditions.

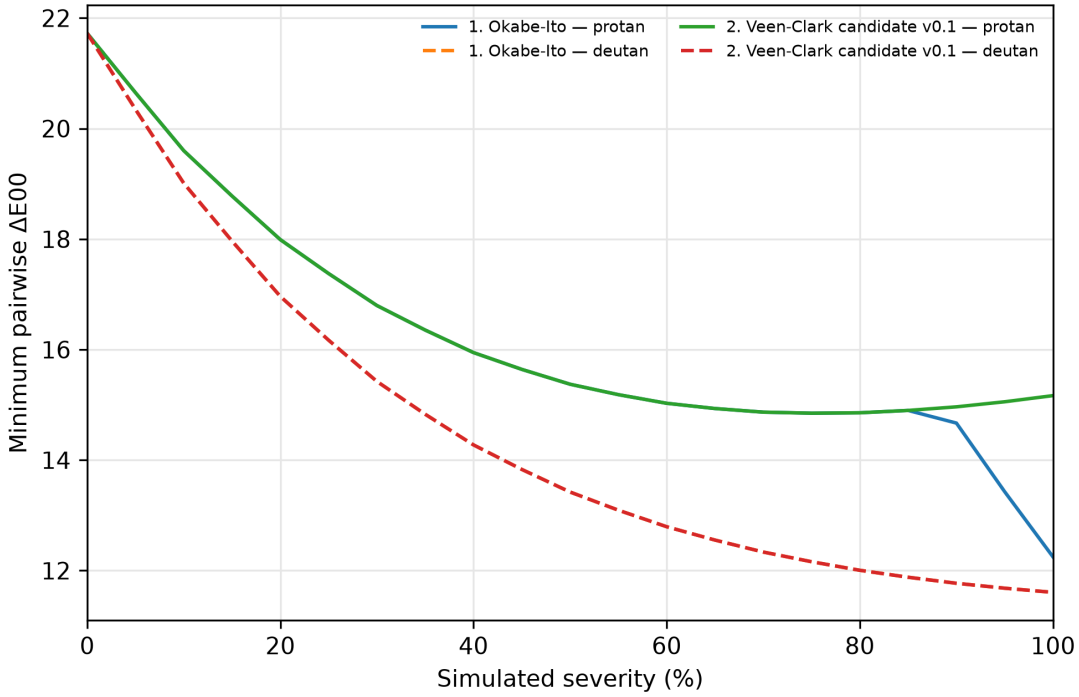


Figure 3. Whole-palette minimum Delta E00 by condition.

Baseline-relative comparison

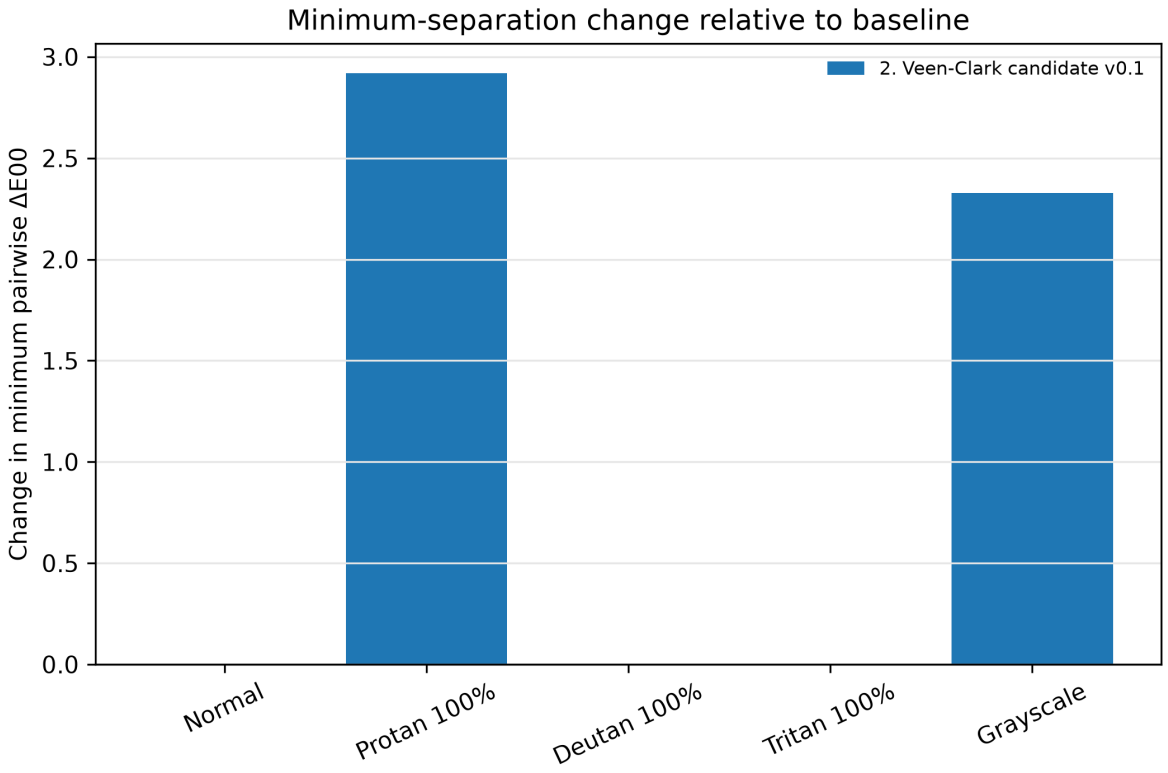


Figure 4. Change in minimum separation relative to Okabe-Ito.

Positive changes indicate greater modelled separation; negative changes indicate reduced modelled separation. No condition is collapsed into an overall score.

Methods and assumptions

Six-digit values are interpreted as gamma-encoded sRGB. PaletteBench applies Colorspacious sRGB1+CVD simulations based on Machado, Oliveira and Fernandes, clips results to displayable sRGB, converts through CIE XYZ to CIE Lab under the sRGB D65 white point, and calculates CIEDE2000 colour differences with Colour Science for Python. Grayscale preserves WCAG relative luminance.

Severity percentages are simulation parameters, not clinical measurements. Partial tritan severity is omitted from the standard report. Clipping provenance and raw values are retained in the machine-readable audit directory.

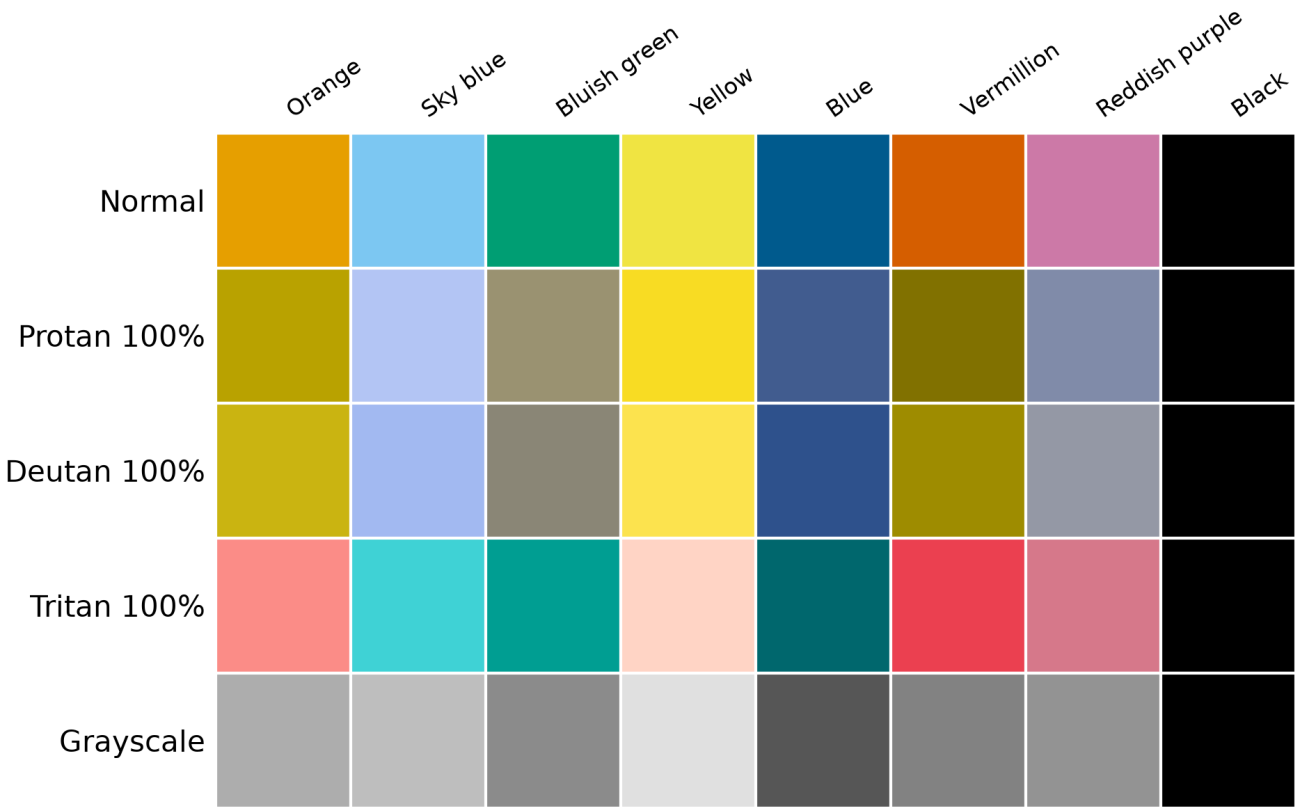


Figure 5. Normal, simulated CVD, and grayscale overview.

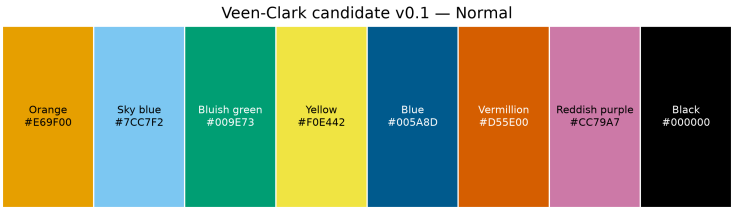
Palette characteristics

Colour	Hex	RGB	L*	Chroma	Luminance	Group
Orange	#E69F00	230 159 0	70.61	76.48	0.4162	-
Sky blue	#7CC7F2	124 199 242	77.01	30.81	0.5154	blue_family
Bluish green	#009E73	0 158 115	57.74	47.49	0.2569	-
Yellow	#F0E442	240 228 66	89.11	76.63	0.7441	-
Blue	#005A8D	0 90 141	36.43	34.62	0.0924	blue_family
Vermillion	#D55E00	213 94 0	54.19	76.67	0.2215	-
Reddish purple	#CC79A7	204 121 167	61.05	39.97	0.2930	-
Black	#000000	0 0 0	0.00	0.00	0.0000	-

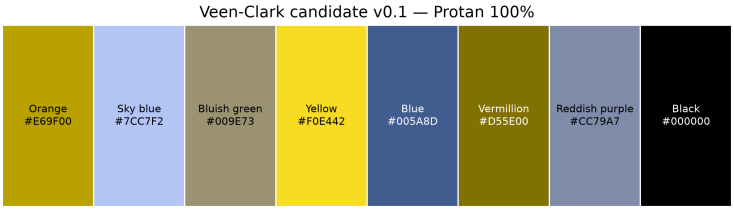
Condition-level summary

Condition	Min	Weakest pair	Mean	Median	10th pct.
normal	21.73	orange / yellow	49.30	49.65	33.71
protan20	17.99	orange / yellow	46.61	47.83	33.91
protan40	15.95	orange / yellow	44.72	44.26	28.67
protan60	15.03	orange / yellow	42.87	42.95	23.52
protan80	14.86	orange / yellow	41.86	42.61	19.46
protan100	15.17	orange / yellow	41.14	41.46	18.08
deutan20	16.96	orange / yellow	46.50	47.93	33.41
deutan40	14.27	orange / yellow	44.72	42.83	29.15
deutan60	12.79	orange / yellow	42.99	40.27	23.95
deutan80	12.00	orange / yellow	41.87	38.78	20.38
deutan100	11.61	orange / yellow	41.36	37.53	16.01
tritan100	10.87	orange / reddish_purple	44.52	52.05	15.79
grayscale	2.93	bluish_green / reddish_purple	26.08	20.65	5.79

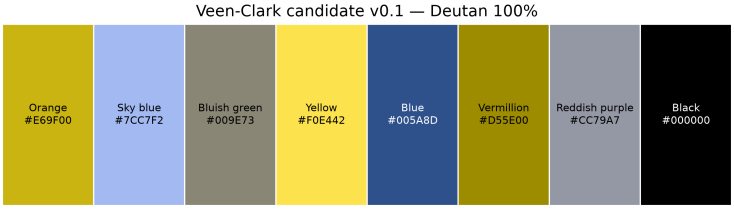
Palette strips



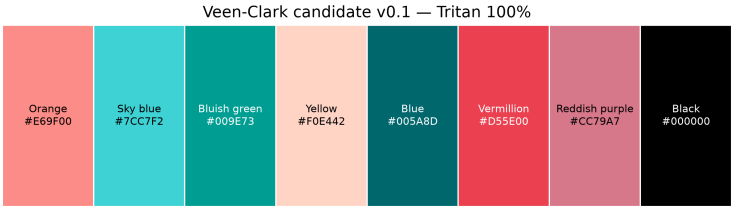
Normal sRGB



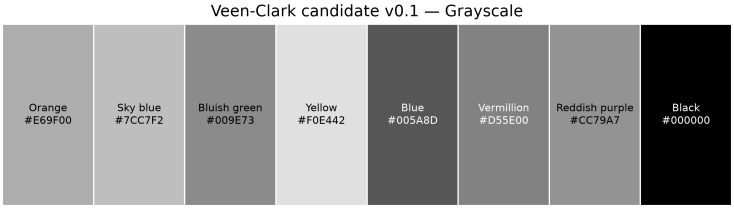
Simulated protan 100%



Simulated deutan 100%



Simulated tritan 100%



WCAG-luminance grayscale

Severity progressions

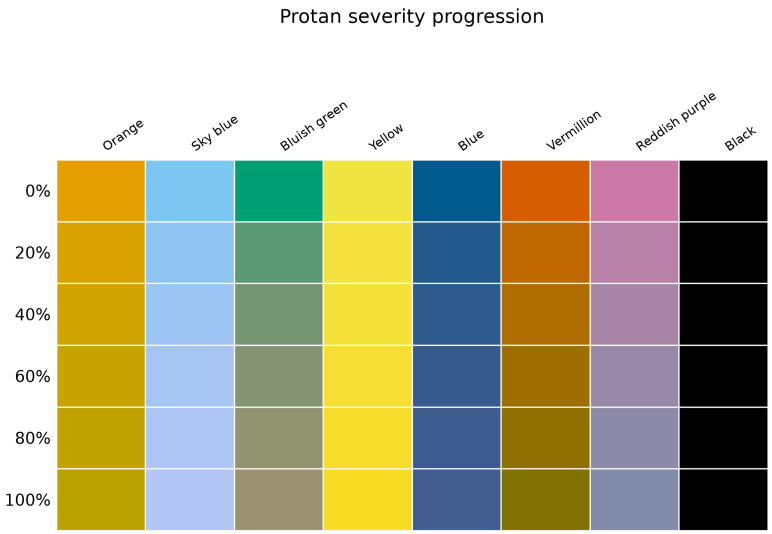


Figure 6. Protan simulation progression from 20% to 100%.

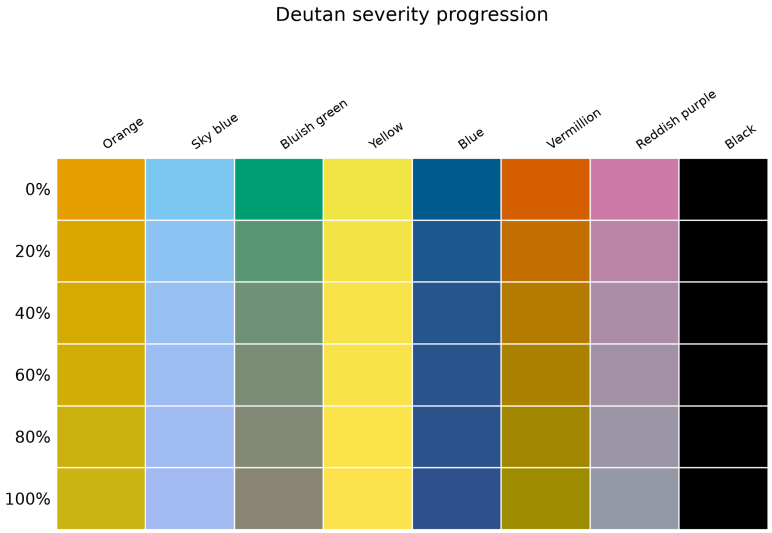


Figure 7. Deutan simulation progression from 20% to 100%.

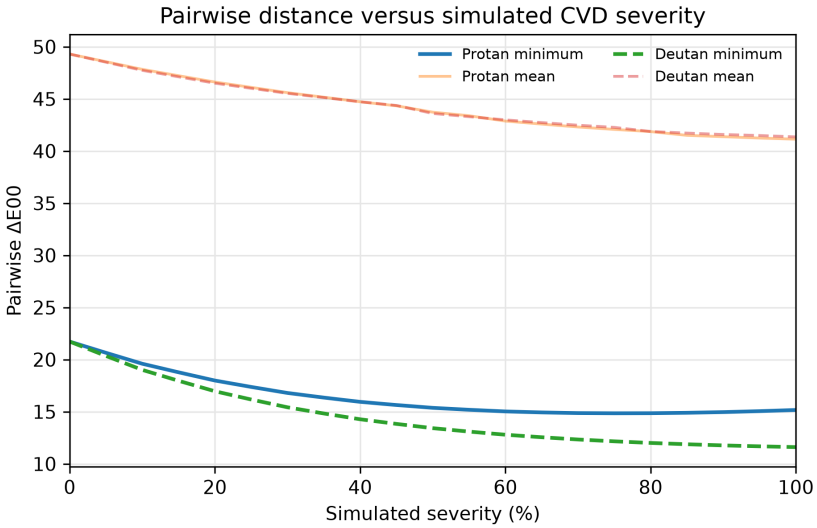
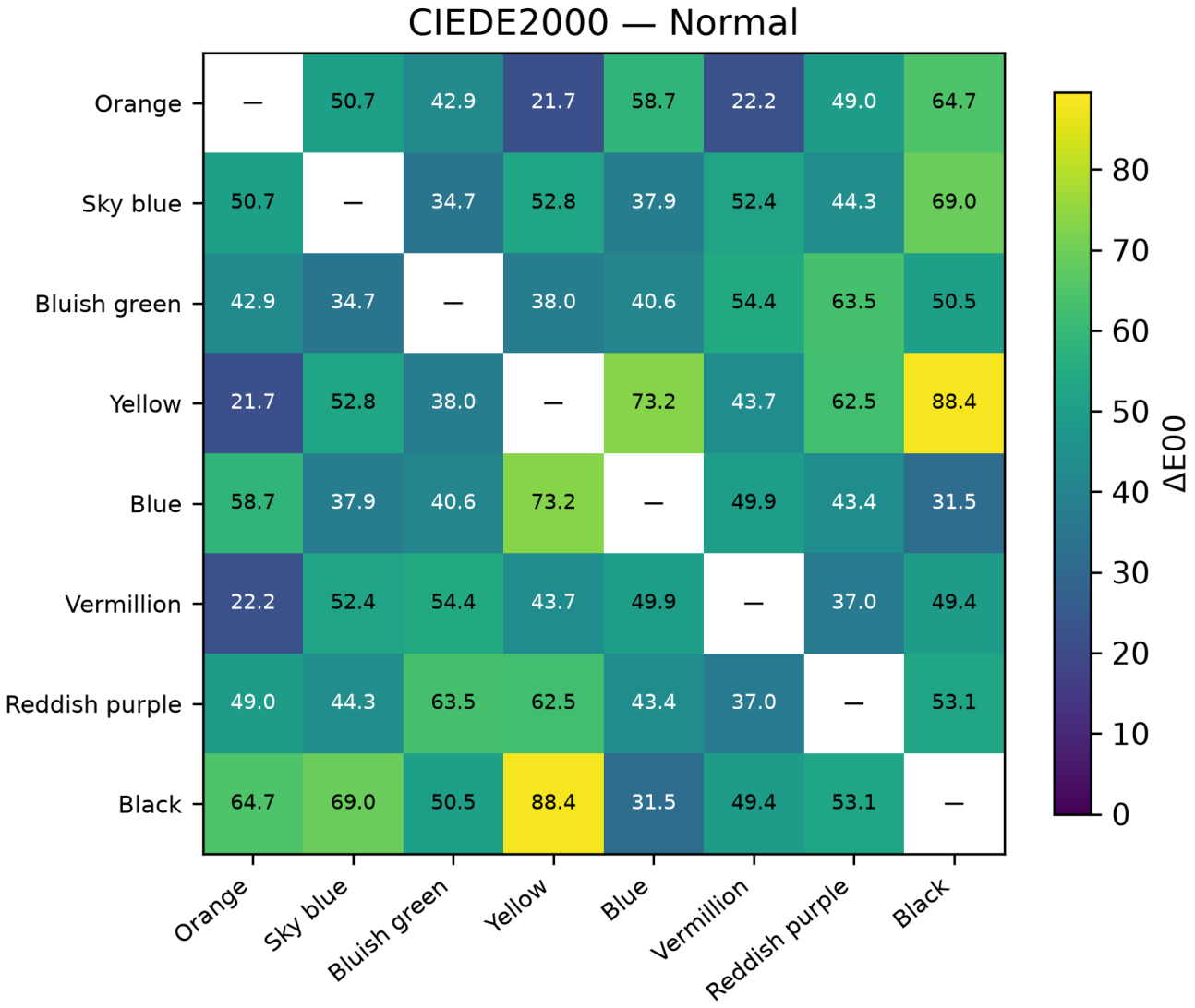


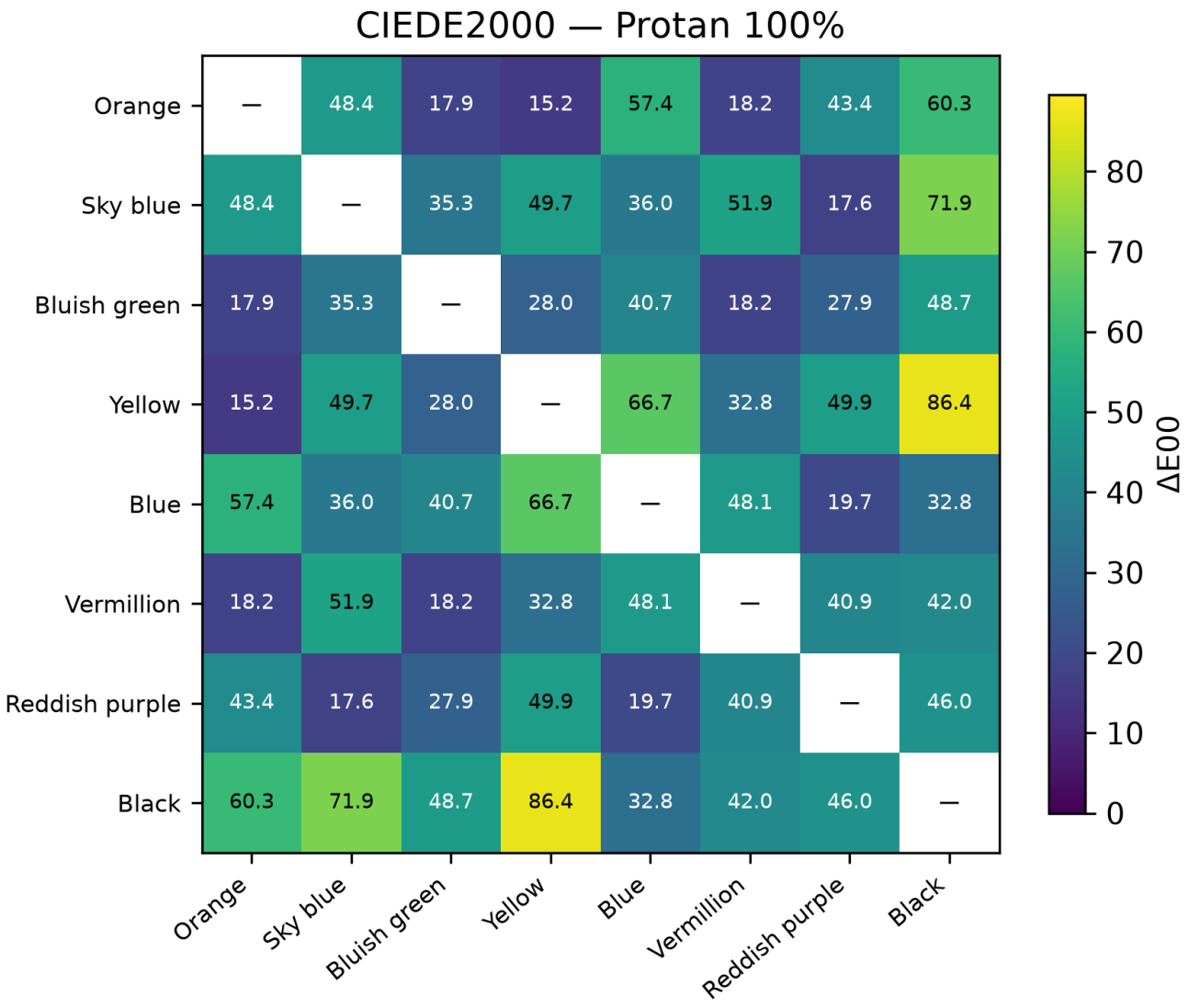
Figure 8. Minimum pairwise distance across simulated severity.

Normal pairwise heatmap



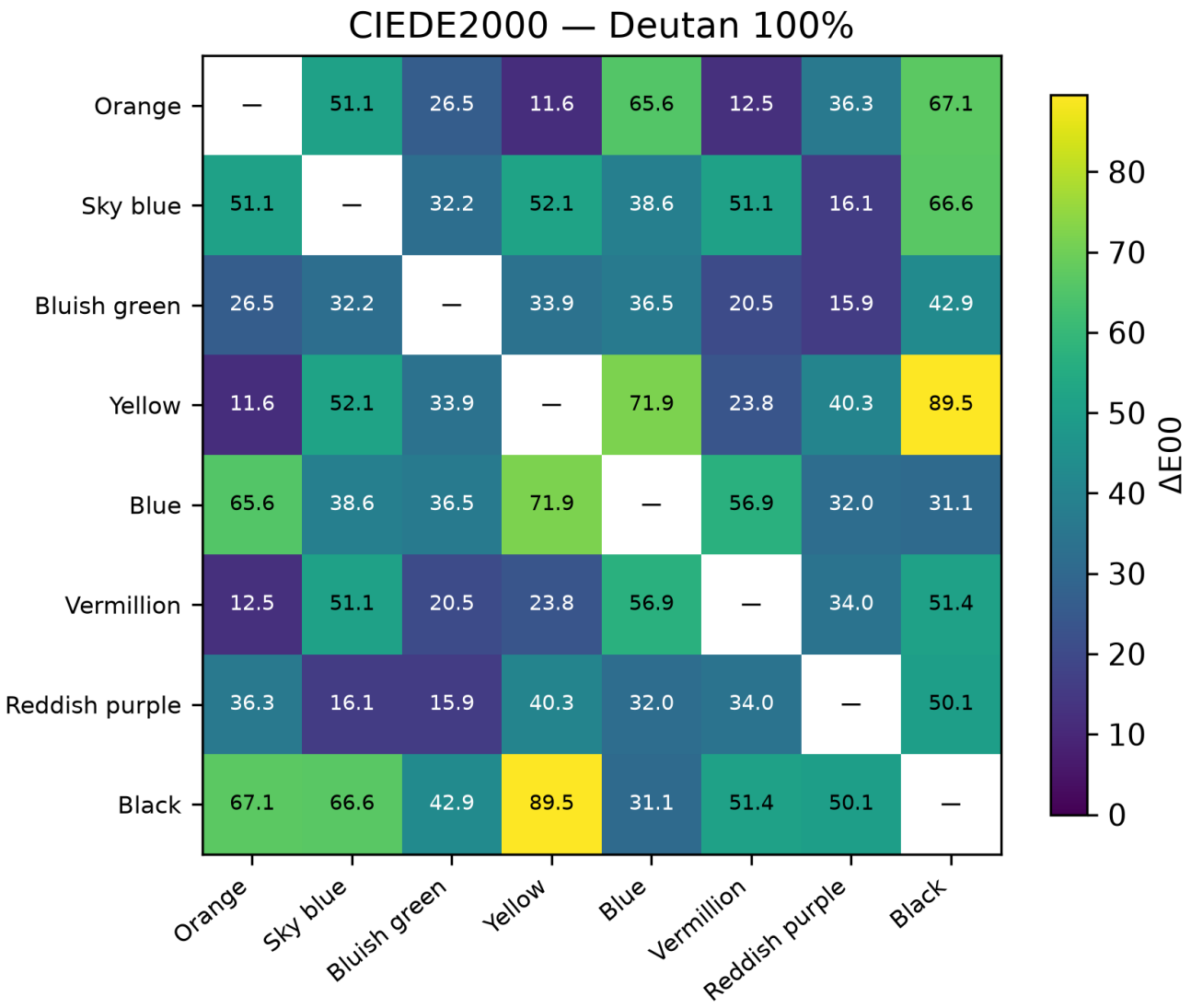
Normal pairwise heatmap.

Protan 100% pairwise heatmap



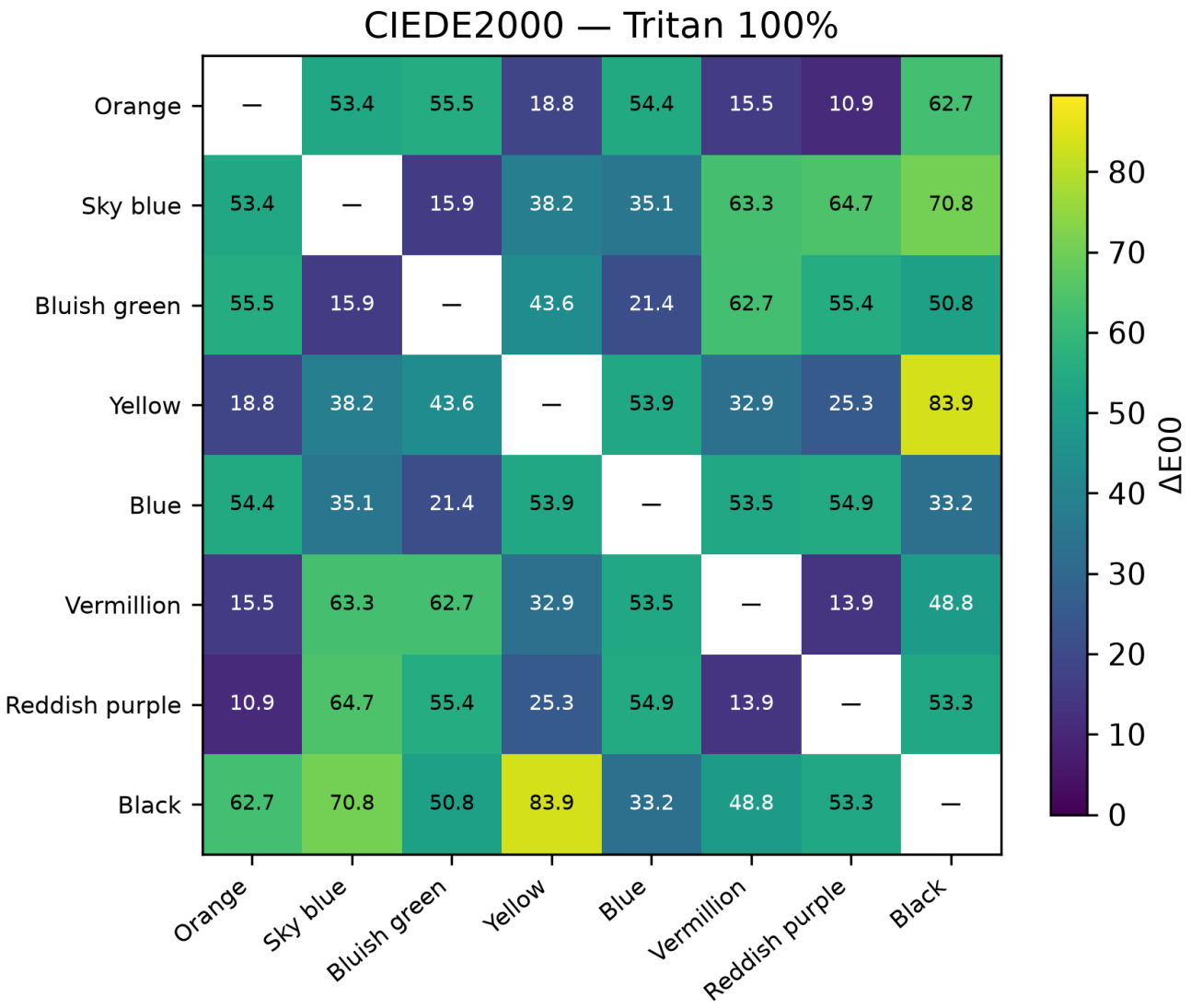
Protan 100% pairwise heatmap.

Deutan 100% pairwise heatmap



Deutan 100% pairwise heatmap.

Tritan 100% pairwise heatmap



Tritan 100% pairwise heatmap.

Normal diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Normal diagonal split pair matrix.

Protan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Protan 100% diagonal split pair matrix.

Deutan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Deutan 100% diagonal split pair matrix.

Tritan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Tritan 100% diagonal split pair matrix.

Grouped blue-family analysis

The YAML declares sky blue and blue as one visual family. With two members, the within-group distribution contains their single pair and directly tracks the intended hierarchical encoding.

Condition	Severity	Pairs	Minimum	Mean	Maximum
normal	0	1	37.92	37.92	37.92
protan20	20	1	37.32	37.32	37.32
protan40	40	1	36.87	36.87	36.87
protan60	60	1	36.53	36.53	36.53
protan80	80	1	36.26	36.26	36.26
protan100	100	1	36.03	36.03	36.03
deutan20	20	1	37.83	37.83	37.83
deutan40	40	1	37.92	37.92	37.92
deutan60	60	1	38.10	38.10	38.10
deutan80	80	1	38.32	38.32	38.32
deutan100	100	1	38.57	38.57	38.57
tritan100	100	1	35.08	35.08	35.08
grayscale	0	1	37.43	37.43	37.43

Weakest-pair diagnostic



Figure 9. Weakest pairs under each standard condition.

Interpretation and limitations

The candidate improves its intended blue-family separation in all standard PaletteBench conditions. It does not improve every pair. In particular, the darker blue approaches black, and its tritan-simulated relationship with vermillion loses separation relative to Okabe-Ito. Redundant encodings remain appropriate whenever category identity is critical.

CVD simulations and perceptual-distance models do not reproduce a particular person's perception. Monitor calibration, viewing environment, acuity, ageing, print reproduction, surrounding colours, spatial context, and mark size can all change discriminability. Human evaluation is outside this candidate audit.

Reproducibility

Palette input SHA-256: 77fc0055ff62d9647bc69a6bff5b098ad78c26d5818be671172d3f5aebd0c1e7. PaletteBench version: 0.1.0. Metric: CIEDE2000 via colour-science; sRGB D65 to CIE Lab. Simulation: Colorspacious sRGB1+CVD (Machado et al.) with display-bound clipping.

References

Okabe, M. and Ito, K. (2002). Color Universal Design: How to make figures and presentations that are friendly to colorblind people.

Wong, B. (2011). Points of view: Color blindness. Nature Methods 8, 441. doi:10.1038/nmeth.1618; corrected 2023, doi:10.1038/s41592-023-01974-0.

Veen, R. J. (2026). PaletteBench 0.1.0. Zenodo. doi:10.5281/zenodo.23071831.